25/04/2025

CODE-A

Aakash

Medical | IIT-JEE | Foundations

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AIM – 720

(Advanced INTENSIVE Mastery for 720)

MM : 720

CST-II

Time : 180 Mins.

Answers

1. (3)	37. (3)	73. (2)	109. (1)	145. (2)
2. (2)	38. (1)	74. (2)	110. (1)	146. (3)
3. (2)	39. (2)	75. (3)	111. (3)	147. (1)
4. (3)	40. (4)	76. (1)	112. (3)	148. (1)
5. (1)	41. (1)	77. (2)	113. (2)	149. (3)
6. (1)	42. (4)	78. (3)	114. (3)	150. (4)
7. (3)	43. (3)	79. (1)	115. (2)	151. (1)
8. (1)	44. (1)	80. (3)	116. (4)	152. (3)
9. (4)	45. (4)	81. (3)	117. (2)	153. (2)
10. (2)	46. (1)	82. (4)	118. (3)	154. (2)
11. (3)	47. (3)	83. (2)	119. (4)	155. (3)
12. (3)	48. (3)	84. (4)	120. (1)	156. (2)
13. (1)	49. (3)	85. (4)	121. (1)	157. (2)
14. (4)	50. (3)	86. (3)	122. (1)	158. (1)
15. (4)	51. (4)	87. (3)	123. (1)	159. (3)
16. (4)	52. (2)	88. (4)	124. (2)	160. (4)
17. (3)	53. (3)	89. (2)	125. (1)	161. (2)
18. (4)	54. (2)	90. (1)	126. (2)	162. (2)
19. (4)	55. (3)	91. (1)	127. (1)	163. (2)
20. (3)	56. (3)	92. (1)	128. (2)	164. (1)
21. (2)	57. (4)	93. (3)	129. (1)	165. (1)
22. (4)	58. (3)	94. (2)	130. (4)	166. (4)
23. (2)	59. (2)	95. (4)	131. (4)	167. (3)
24. (3)	60. (4)	96. (3)	132. (1)	168. (1)
25. (4)	61. (4)	97. (1)	133. (1)	169. (2)
26. (3)	62. (4)	98. (2)	134. (4)	170. (2)
27. (2)	63. (4)	99. (1)	135. (1)	171. (4)
28. (2)	64. (2)	100. (2)	136. (2)	172. (1)
29. (2)	65. (2)	101. (3)	137. (2)	173. (4)
30. (3)	66. (2)	102. (3)	138. (1)	174. (2)
31. (4)	67. (1)	103. (3)	139. (2)	175. (4)
32. (4)	68. (4)	104. (3)	140. (2)	176. (1)
33. (4)	69. (2)	105. (3)	141. (2)	177. (3)
34. (1)	70. (3)	106. (2)	142. (1)	178. (1)
35. (4)	71. (4)	107. (2)	143. (4)	179. (4)
36. (3)	72. (4)	108. (1)	144. (3)	180. (4)

Hints and Solutions

BIOLOGY

(1) **Answer :** (3)
Solution:
Flowers in mango are bisexual. Ethylene initates flowering in mango.

(2) **Answer :** (2)
Solution:
Increased vacuolation is seen in the cells of elongation phase of the plant parts such as root tips.

(3) **Answer :** (2)
Solution:
Cells of tapetum possess dense cytoplasm and generally have more than one nucleus. They are polyploid.

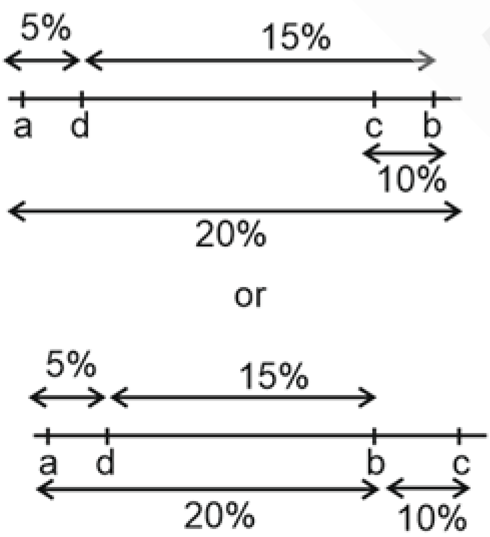
(4) **Answer :** (3)
Solution:
A fruit which develops from other floral parts along with the development of ovary wall is called false fruit.

(5) **Answer :** (1)
Solution:
A typical angiosperm embryo sac shows the presence of 7-cells that are, one egg cell, two synergids, three antipodals and one central cell. But since, the central cell has two polar nuclei, the embryo sac is 8-nucleate and 7-celled at maturity.

(6) **Answer :** (1)
Solution:
Myotonic dystrophy is an example of autosomal dominant disorder.

(7) **Answer :** (3)
Solution:
Sickle-cell anaemia is an autosome linked recessive trait that is caused by the substitution of glutamic acid by valine at the sixth position of the β-globin chain of haemoglobin.

(8) **Answer :** (1)
Solution:
The possible arrangement of the genes can be



(9) **Answer :** (4)
Solution:
Dominance is not an autonomous feature of a gene. It depends much on the product of the gene and the production of a particular phenotype from this product.

(10) **Answer :** (2)
Solution:

In a eukaryotic cell, mRNA must be processed before it is translated. Polyadenylation, capping and splicing are the steps involved in mRNA processing.

(11) **Answer :** (3)
Solution:
The anticodon on tRNA for the codon 5′ AGU 3′ will be 5’ ACU 3’

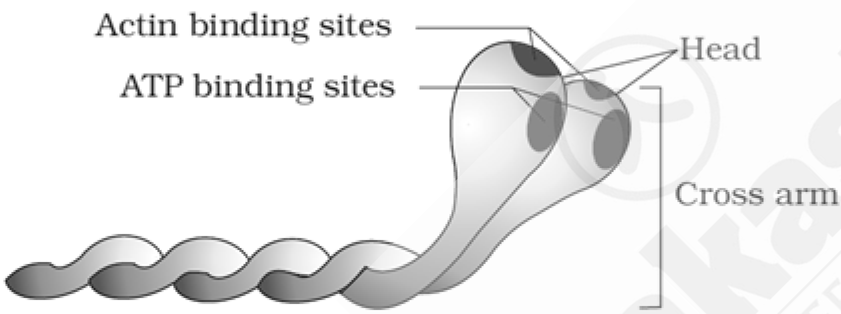
(12) **Answer :** (3)
Solution:
Promoter is a DNA sequence that provides binding site for RNA polymerase.

(13) **Answer :** (1)
Solution:
The mutation in operator region of operon leads to the constitutive expression of *lac* operon as the repressor cannot bind to the operator region.

(14) **Answer :** (4)
Solution:
In dicot roots, conjunctive tissue and pericycle are found.

(15) **Answer :** (4)
Solution:
Casparian strips are found in the endodermis of roots. Water molecules are unable to penetrate this layer in the areas of casparian strips.

(16) **Answer :** (4)
Solution:
A globular head with a short arm is called heavy meromyosin.



(17) **Answer :** (3)
Solution:
In a polymerase chain reaction, the number of DNA strands doubles with each cycle, following the formula:
Number of copies of DNA strands = 2^n.
Here ‘n’ is the number of cycles of PCR.
Thus, to generate 256 DNA strands from a single dsDNA piece, 8 PCR cycles are required.

(18) **Answer :** (4)
Solution:
*Eco*RI produces sticky ends by cutting the strand of DNA a little away from the centre of the palindromic sites but between the same two bases on opposite strands.

(19) **Answer :** (4)
Solution:
In all connective tissues, except blood and lymph, the cells secrete fibres of structural proteins called collagen or elastin.

(20) **Answer :** (3)
Solution:
Three major regions make up the brain stem, *i.e.*, midbrain, pons and medulla oblongata.
Amygdala, hippocampus and hypothalamus are the parts of limbic system in forebrain.

(21) **Answer :** (2)
Solution:
The DNA fragments separate (resolve) according to their size through sieving effect provided by the agarose gel.
The construction of the first recombinant DNA emerged from the possibility of linking a gene encoding antibiotic resistance with a native plasmid (autonomously replicating circular extra-chromosomal DNA) of *Salmonella typhimurium*.

(22) **Answer :** (4)
Solution:

The property common to both nerve cells (neurons) and muscle cells (myocytes) is excitability.
Excitability refers to the ability of a cell to respond to stimuli by generating and propagating electrical impulses. Both neurons and muscle cells possess excitable membranes that can produce electrochemical impulses and conduct them along their surfaces.
The other properties listed-contractility, elasticity, and extensibility are primarily associated with muscle cells and are not the characteristics of nerve cells. Contractility is the ability of muscle cells to shorten and generate force. Elasticity refers to their ability to return to original length after stretching and extensibility is the capacity to be stretched without damage.

(23) **Answer :** (2)
Solution:
SAN is present in right upper corner of right atrium and AVN is present in lower left corner of right atrium. A thin, muscular wall called the inter-atrial septum seperates right and left atrium. The atrium and the ventricle of the same side are also seperated by a thick fibrous tissue called the atrio-ventricular septum.

(24) **Answer :** (3)
Solution:
Intercellular material of cartilage is solid and pliable and resists compression. Chondrocytes are enclosed in small cavities within matrix secreted by them.

(25) **Answer :** (4)
Solution:
Columba and *Pavo* belong to the class Aves so they have double circulation. *Ornithorhynchus* belongs to class Mammalia. It also has four-chambered heart and exhibits double circulation. *Carcharodon* belongs to the class Chondrichthyes and has two-chambered heart and exhibits single circulation.

(26) **Answer :** (3)
Solution:
Lateral meristems are generally not present from the very beginning of life of a plant and appear later than primary meristems. These are cylindrical meristems which result in increase in the girth of stems and roots. Apical meristems are found at the apices of stems and roots.

(27) **Answer :** (2)
Solution:
Nucleoid is the packaged structure of bacterial DNA. Clover-leaf is a secondary structure of tRNA. Regulator gene transcribes mRNA for protein synthesis.

(28) **Answer :** (2)
Solution:
Middle lamella is chiefly made up of calcium and magnesium pectate.

(29) **Answer :** (2)
Solution:
Plasmodesmata - Cytoplasmic connections
Mitochondria – Site for the synthesis of ATP
Axoneme - Core of cilia and flagella
Nucleolus - Site for rRNA synthesis

(30) **Answer :** (3)
Solution:
No organelle duplication takes place in the M phase of cell cycle.

(31) **Answer :** (4)
Solution:
Tetrad of cells is formed during telophase II.

(32) **Answer :** (4)
Solution:
Intermediate filaments form scaffolds for chromatin.

(33) **Answer :** (4)
Solution:
Mitosis helps in maintaining the cell size as it induces a cell to divide whenever the surface to volume ratio is disturbed, as the cell grows in size.

(34) **Answer :** (1)
Solution:
In both, green algae and brown algae, asexual reproduction is by flagellated zoospores. Red algae reproduce asexually by non-motile spores. Red algae are able to grow in great depth and they store food in the form of floridean starch and glycogen.

(35) Answer : (4)
Solution:
In *Marchantia,* male and female sex organs are present on different thalli, whereas' in *Funaria* both male and female sex organs are present on the same thalli.

(36) Answer : (3)
Solution:
Lactobacillus is used for the commercial production of lactic acid.

(37) Answer : (3)
Solution:
Biological control methods adopted in agricultural pest control are based on the ability of the predator to regulate the prey population.

(38) Answer : (1)
Solution:
Chanda and Bastar areas are in Madhya Pradesh.

(39) Answer : (2)
Solution:
In co-extinction, when a species becomes extinct, the plant and animal species associated with it, in an obligatory way, also become extinct.

(40) Answer : (4)
Solution:
In the members of Phaeophyceae, the plant body is usually attached to the substratum by a holdfast and has a stalk, the stipe and leaf like photosynthetic organ, the frond. *Laminaria* belongs to Phaeophyceae.

(41) Answer : (1)
Solution:
Fibrinogens are needed for clotting or coagulation of blood. Globulins are primarily involved in defense mechanism of the body and the albumins help in osmotic balance. GLUT-4 enables glucose transport into cells.

(42) Answer : (4)
Solution:
The cutting of DNA at specific locations became possible with the discovery of the so-called ‘molecular scissors’- restriction enzymes.
The linking of antibiotic resistance gene with the plasmid vector became possible with the enzyme DNA ligase, which acts on cut DNA molecules and joins their ends.

(43) Answer : (3)
Solution:
Red muscle fibres have large amount of myoglobin and mitochondria while white muscle fibres have large amount of sarcoplasmic reticulum and depend on anaerobic process for energy.

(44) Answer : (1)
Solution:
In alveoli, where there is low pCO_2, high pO_2, lesser H^+ concentration and lower temperature, the factors are all favourable for the formation of oxyhaemoglobin.

(45) Answer : (4)
Solution:
When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibitor *e.g*., Inhibition of succinic dehydrogenase by malonate which closely resembles the substrate succinate in the structure.

(46) Answer : (1)
Solution:
In peroxidase and catalase, which catalyze the breakdown of hydrogen peroxide to water and oxygen, haem is the prosthetic group and it is a part of the active site of the enzyme.

(47) Answer : (3)
Solution:
The squamous epithelium is made of a single layer of flattened cells with irregular boundaries. They are found in the walls of blood vessels and air sacs of lungs and are involved in functions like forming a diffusion boundary. Squamous epithelium does not possess microvilli.

(48) Answer : (3)
Solution:

Enzyme catalysts differ from inorganic catalysts in many ways. Inorganic catalysts work efficiently at high temperatures and high pressures, while enzymes get damaged at high temperatures (say above 40°C). However, enzymes isolated from organisms who normally live under extremely high temperatures (*e.g*., hot vents and sulphur springs), are stable and retain their catalytic power even at high temperatures (upto 80°-90°C).

(49) Answer : (3)
Solution:
Living organisms have a number of carbon compounds in which heterocyclic rings can be found. Some of these are nitrogenous bases - adenine, guanine, cytosine, uracil, and thymine.

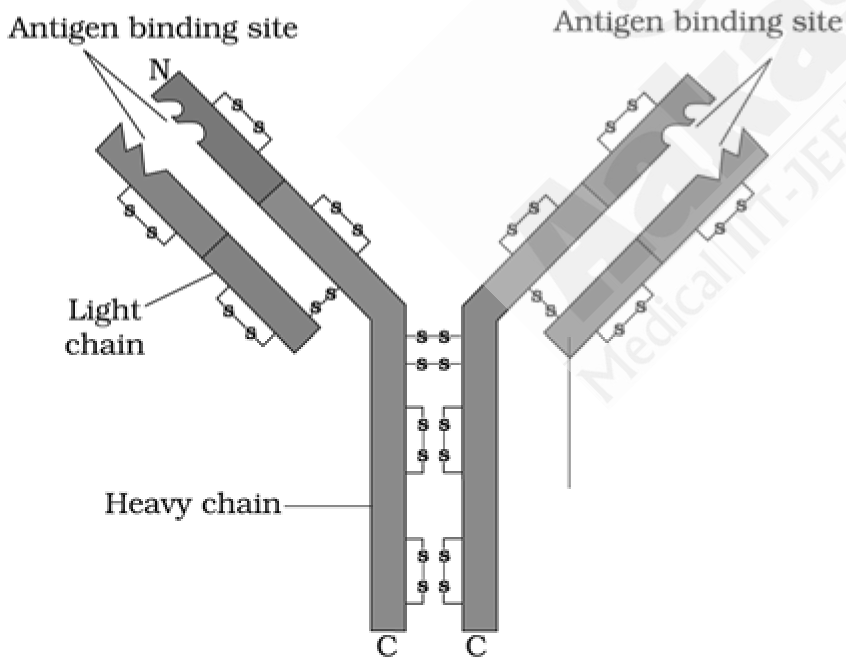
(50) Answer : (3)
Solution:
During expiration, relaxation of the diaphragm and the external inter-costal muscles leads to an increase in intra-pulmonary pressure to slightly above the atmospheric pressure causing the expulsion of air from lungs.

(51) Answer : (4)
Solution:
Pituitary, pineal, thyroid, adrenal, pancreas, parathyroid, thymus and gonads (testes in males and ovaries in females) are the organised endocrine bodies in our body.

(52) Answer : (2)
Solution:
Testes, ovaries and adrenal glands are paired structures.
Testes produce a group of hormones called androgens mainly testosterone.
Ovaries produce two steroid hormones called estrogen and progesterone.
Adrenal glands (adrenal cortex) release cortisol, androgenic steroids, mineralocorticoids, *etc*.

(53) Answer : (3)
Solution:
Here, ‘X’ refers to thymus which is a primary lymphoid organ, It is an unpaired gland and secretes peptide hormone called thymosin.

(54) Answer : (2)
Solution:



(55) Answer : (3)
Solution:
Active immunity is slow and takes time to give its full effective response.
Injecting the microbes deliberately during immunisation or infectious organisms gaining access into body during natural infection induce active immunity.

(56) Answer : (3)
Solution:
Acromegaly is caused due to hypersecretion of growth hormone during adulthood where as Graves' disease is caused due to hypersecretion of tetraiodothyronine (thyroxine).

(57) Answer : (4)
Solution:
Presence of notochord at some stage of their life is fundamental feature of all chordates. *Catla* is a fresh water bony fish which possesses air bladder that provides buoyancy. *Myxine* is a cyclostome in which scales are absent. Cycloid/ctenoid scales are present in *Catla*.

(58) Answer : (3)
Solution:
Under normal physiological conditions, the filtrate formed per day is 180 L and urine released per day is 1-1.5 L.

Hence, ratio between these two will be $\frac{180\text{ L/day}}{1.5\text{ L/day}} = 120 : 1$

(59) Answer : (2)
Solution:
Tyrannosaurus rex was 20 feet in height and had huge fearsome dagger-like teeth. *Pteranodon* was a flying reptile.

(60) Answer : (4)
Solution:
Even before Darwin, a French naturalist Lamarck had said that evolution of life forms had occurred but driven by use and disuse of organs. Wallace worked in Malay Archipelago. Hugo deVries gave the idea of mutation.

(61) Answer : (4)
Solution:
In the given figure 'A' represents bobcat and 'B' represents wolf. Both bobcat and wolf are placental mammals and exhibits convergent evolution with tasmanian tiger cat and tasmanian wolf respectively.

(62) Answer : (4)
Solution:
Conditional reabsorption of Na^+ and water takes place in DCT. It is also capable of reabsorption of HCO_3^- and selective secretion of hydrogen and potassium ions and NH_3 to maintain the pH and sodium-potassium balance in blood.

(63) Answer : (4)
Solution:
Limulus (King crab) possesses book gills for respiration.

(64) Answer : (2)
Solution:
Process in which isolated protoplasts from two different varieties of plants-each having desirable character can be fused to get hybrid plants is called somatic hybridisation.

(65) Answer : (2)
Solution:
Disarmed *Agrobacterium* fumefaciens vector is used to introduce nematode specific genes into the host plant to initiate RNAi.

(66) Answer : (2)
Solution:
Camembert cheese - *Penicillium camembertii*
Swiss Cheese - *Propionibacterium sharmanii*
Dosa - *Leuconostoc*
Pectinase - *Aspergillus niger*

(67) Answer : (1)
Solution:
In the given graph, curve ‘a’ is described by $\frac{dN}{dt} = rN$, and ‘b’ is described by $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$

(68) Answer : (4)
Solution:
The special anatomy of leaves of the C_4 plant is called Kranz anatomy. Bell pepper is a C_3 plant. Both C_3 and C_4 plants utilize 2 molecules of NADPH per CO_2 fixed.

(69) Answer : (2)
Solution:
Non-cyclic photophosphorylation occurs in the membrane of granal thylakoids.

(70) Answer : (3)
Solution:
In TCA cycle, the first decarboxylation reaction occurs during the conversion of oxalosuccinic acid to α-ketoglutaric acid.

(71) Answer : (4)
Solution:
According to 10% law, only 10% of the energy is transferred to each trophic level from the lower trophic level. Incident solar radiation energy 200,000 J. Only 1% of energy is trapped at T_1 level. Therefore, at T_1 (2000 J) of energy is present.

T_1 →	T_2 →	T_3 →	T_4
(2000 J)	(200 J)	(20 J)	(2 J)
Energy utilised	T_1	T_2	T_3
	1800 J	180 J	18 J

(72) Answer : (4)
Solution:
In alcoholic fermentation, acetaldehyde is reduced to alcohol by NADH + H^+ by the action of enzyme alcohol dehydrogenase.

(73) Answer : (2)
Solution:
The rate of decomposition is slower if the detritus is rich in lignin and chitin.

(74) Answer : (2)
Solution:
Genus comprises a group of related species which has more characters in common in comparison to species of other genera.

(75) Answer : (3)
Solution:
Neurospora is used extensively in biochemical and genetic work.

(76) Answer : (1)
Solution:
Imbricate aestivation is seen in *Cassia* and Gulmohur.

(77) Answer : (2)
Solution:
The flowers of Liliaceae family are bisexual and actinomorphic.

(78) Answer : (3)
Solution:
Sunflower and marigold show basal placentation.

(79) Answer : (1)
Solution:
In *Agaricus* and *Puccinia,* the mycelium is branched and septate.

(80) Answer : (3)
Solution:
Viruses are non-cellular organisms that are characterised by having an inert crystalline structure outside the living cell.

(81) Answer : (3)
Solution:
In ELISA, infection by pathogen can be detected by the presence of antigens (proteins, glycoproteins, *etc*.) or by detecting the antibodies synthesised against the pathogen.

(82) Answer : (4)
Solution:
Cuttlefish - Presence of radula
Prawn - Statocyst for balancing
Scorpion - Possesses book lungs for respiration
Roundworm - Dioecious in nature

(83) Answer : (2)
Solution:
In male frog, testes are found adhered to upper part of kidneys by a double fold of peritoneum called mesorchium.
Endometrium is the innermost and myometrium is the middle layer present in the wall of human uterus.

(84) Answer : (4)

Solution:
The regions in between seminiferous tubules are called interstitial spaces which contain interstitial cells, immunologically competent cells and small blood vessels. Sertoli cells are present in the inner lining of seminiferous tubules.

(85) Answer : (4)
Solution:
Formation of first polar body occurs in tertiary follicle in ovaries. While second polar body is formed in in fallopian tubes during fertilisation.

(86) Answer : (3)
Solution:
Prostate gland is an unpaired gland while epididymis is an accessory duct.

(87) Answer : (3)
Solution:
In male cockroaches, the genital pouch lies at the hind end of abdomen bounded dorsally by 9^{th} and 10^{th} terga and ventrally by 9^{th} sternum.

(88) Answer : (4)
Solution:
The sperm head contains an elongated nucleus. The anterior portion of the head is covered by a cap-like structure, acrosome. A plasma membrane envelops the whole body of sperm.

(89) Answer : (2)
Solution:
Oral contraceptive pills containing progestogens alone or progesterone-estrogen in combination, inhibit ovulation and implantation as well as alter the quality of cervical mucus to prevent/retard entry of sperms.

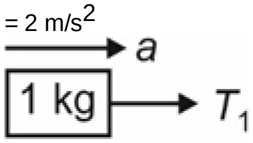
(90) Answer : (1)
Solution:
'Nirodh' is a popular brand of condom for the males. Use of condoms has increased in recent years due to its additional benefit of protecting the user from contracting STIs and AIDS.
Copper releasing IUDs - Multiload 375, Cu7
Hormone releasing IUDs - LNG-20
Copper released in copper releasing IUDs suppresses sperm motility and the fertilising capacity of sperms.
In tubectomy, a small part of the fallopian tube is removed or tied up through a small incision in the abdomen or through vagina, this technique is highly effective but its reversibility is very poor.

PHYSICS

(91) Answer : (1)
Solution:
A neutral body is charged positively when it loses electrons and hence the mass of the body will decrease.

(92) Answer : (1)
Solution:
$E = -L\frac{di}{dt}$
$E = 0.02 \times \frac{4}{2\times 10^{-2}}$
= 4 V

(93) Answer : (3)
Solution:
Acceleration of system (a) = $\frac{F_{net}}{m}$
$= \frac{16}{8}$
= 2 m/s^2



T_1 = 1×2
T_1 = 2 N

(94) Answer : (2)

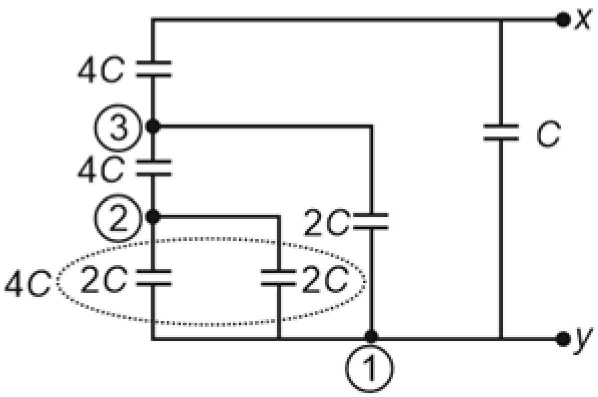
Solution:

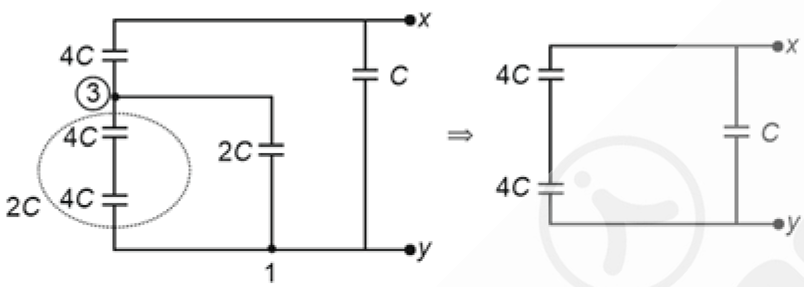
$V = \frac{kq}{r}$

$V_T = \frac{k}{1}[8 + 5 - 3 - 10] \times 10^{-6} = 0$

(95) Answer : (4)

Solution:





$C_{xy} = 2C + C$

$\quad = 3\,C$

(96) Answer : (3)

Solution:

$i = neAv_d \;\Rightarrow\; i = neA \times v$ ...(i)

$2i = ne(2A) \times v'$ ...(ii)

$v' = v$

(97) Answer : (1)

Solution:

In half defection method,

$G = \frac{RS}{R-S}$

$= \frac{1000\times 20}{1000-20}$

$= \frac{1000\times 20}{980}$

$= \frac{1000}{49}\ \Omega$

(98) Answer : (2)

Solution:

When p-side of diode is connected to higher potential, then diode is in forward biasing.

(99) Answer : (1)

Solution:

$n_A = 10^{18}$ atoms/m^3

$n_i = 10^{16}$ m^{-3}

$n_e = \frac{n_i^2}{n_A} = \frac{10^{32}}{10^{18}} = 10^{14}\,\mathrm{m}^{-3}$

(100) Answer : (2)

Solution:

Energy of incident photon = $\frac{12400}{2000}$ = 6.2 eV
⇒ $(K.E)_{max} = h\nu - \varphi$
$(K.E)_{max} = 6.2 - 4.5$
= 1.7 eV
∴ Stopping potential = – 1.7 V

(101) Answer : (3)
Solution:
Energy of incident photon $E = 8 + 4.09$
= 12.09 eV
⇒ ΔE_{Photon} = 12.09 eV
$= E_n - E_1$
∴ $\frac{-13.6}{n^2} + 13.6$ = 12.09
$n = 3$

(102) Answer : (3)
Solution:
Mass defect Δm = 0.2% of 1 kg
$= \frac{0.2}{100} \times 1$
= 2×10^{-3} kg
Energy released $E = \Delta mc^2$
$= 2\times10^{-3}\times(3\times10^8)^2$
$= 2\times10^{-3} \times 9\times10^{16}$
$= 18\times10^{13}$ J

(103) Answer : (3)
Solution:
m = 1 kg
$W_{net} = \frac{1}{2}m\left(v_f^2 - v_i^2\right)$
$= \frac{1}{2}m\left(\alpha^2 x_f^3 - \alpha^2 x_i^2\right)$
$= \frac{1}{2} \times (2700 - 100)$
= 1300 J

(104) Answer : (3)
Solution:
Here the motion is not accelerated, the resultant force parallel to the inclined plane must be zero.
So, $F - mg\sin30^o = 0$
$F = mg\sin30^o$
$W = Fd\cos\theta$
$= mg\sin30^o d\cos0^o$
= 500 J

(105) Answer : (3)
Solution:
If multiple identical cells are connected in series in same sense, the total EMF increases.

(106) Answer : (2)
Solution:
$F = \frac{vdm}{dt} = 20 \times 2 = 40\ \mathrm{N}$
$a = \frac{F}{m} = \frac{40}{2} = 20\ \mathrm{m}/s^2$

(107) Answer : (2)
Solution:
$v_{avg} = \frac{2x}{\frac{x}{2} + \frac{x}{5}}$
$= \frac{20}{7}\ \mathrm{m}/s$

(108) Answer : (1)
Solution:
Speed at mid point is given by

$v_m = \sqrt{\frac{u^2+v^2}{2}}$

$v_m = \sqrt{\frac{100+400}{2}} = \sqrt{250} = 5\sqrt{10}$ m/s

(109) Answer : (1)

Solution:

$[MLT^{-2}] = [a]$

$\left[\frac{b}{L}\right] = \left[M^0L^0T^0\right]$

$[b] = [L]$

(110) Answer : (1)

Solution:

$V = a^3 = 8\ cm^3$

$\frac{\Delta V}{V} = \frac{3\Delta a}{a} \Rightarrow \Delta V = 3V\left(\frac{\Delta a}{a}\right) = 3\times 8\times\left(\frac{0.01}{2.00}\right)$

$= 0.12\ cm^3$

$V = (8 \pm 0.12)\mathrm{cm}^3$

Surface area = $6a^2 = 6(2.00)^2 = 24.0\ cm^2$

$\frac{\Delta A}{A} = \frac{2\Delta a}{a} \Rightarrow \Delta A = 2A\left(\frac{\Delta a}{a}\right) = 0.24$

$\therefore\ A = (24.0 \pm 0.24)\ \mathrm{cm}^2$

(111) Answer : (3)

Solution:

$x = 10\cos\left(2\pi(1) + \frac{\pi}{4}\right)$

$= 10\cos\left(2\pi + \frac{\pi}{4}\right)$

$= 10\cos\frac{\pi}{4}$

$= 10 \times \frac{1}{\sqrt{2}} = 5\sqrt{2}$ cm

(112) Answer : (3)

Solution:

Velocity of sound $v = \sqrt{\frac{\gamma P}{\rho}}$

$\rho_{\text{moist air}} < \rho_{\text{dry air}}$

Velocity of sound in moist air is more than in dry air.

(113) Answer : (2)

Solution:

For first minimum, path difference $a\sin\theta = \lambda$

$a\sin 30° = \lambda$

$a = 2\times\lambda = 2\lambda$

For first secondary maximum, path difference

$a\sin\theta' = \frac{3}{2}\lambda$

$\sin\theta' = \frac{3\lambda}{2(2\lambda)}$

$\theta' = \sin^{-1}\left(\frac{3}{4}\right)$

(114) Answer : (3)

Solution:

Infrared rays are used in remote switches, microwaves are used in cooking, and UV rays are used to kill germs in water purifiers.

Gamma rays are used in medicine to destroy cancer cells.

(115) Answer : (2)

Solution:

$\Delta l = \frac{FL}{AY}$

$= \frac{(100\times 10^3 \mathrm{N})\times 1\ \mathrm{m}}{3.14\times\left(10^{-2}\mathrm{m}\right)^2\times 2\times 10^{11}\mathrm{N}/m^2}$

= 1.59 mm

(116) Answer : (4)
Solution:
Velocity of object *A* with respect to object *B*

$\vec{v}_{A/B} = \vec{v}_A - \vec{v}_B$

$= 3\hat{i} + 4\hat{j} - (7\hat{i} - 3\hat{j})$

$= 3\hat{i} + 4\hat{j} - 7\hat{i} + 3\hat{j}$

$= \left(-4\hat{i} + 7\hat{j}\right)$ m/s

(117) Answer : (2)
Solution:

Centre of mass of this rigid rod, $x_{cm} = \dfrac{\int x\,dm}{\int dm}$

where $\frac{dm}{dx} = 3x \Rightarrow \; dm = 3x\,dx$

$\therefore \dfrac{\int_0^1 3x^2\,dx}{\int_0^1 3\,x dx} = \dfrac{3\times \frac{x^3}{3}\Big|_0^1}{3\times \frac{x^2}{2}\Big|_0^1} = \dfrac{3\times\frac{1}{3}}{3\times\frac{1}{2}} = \frac{2}{3}$m

(118) Answer : (3)
Solution:
In a perfectly rigid body, there is no internal motion. The work done by external torques is therefore, not dissipated and goes on to increase the rotational kinetic energy of the body.
Torque τ = *I*α ⇒ α is proportional to $\frac{1}{I}$

(119) Answer : (4)
Solution:
The two masses will always remain diametrically opposite to keep revolution around a common centre.

$U = \frac{-2GM^2}{R} - \frac{GM^2}{2R} = \frac{-5GM^2}{2R}$

(120) Answer : (1)
Solution:
Physical balance measures the mass whereas spring balance measures the weight.
Reading on spring balance increases as $g_{\text{pole}} > g_{\text{equator}}$
Reading on physical balance remains the same.

(121) Answer : (1)
Solution:
For green light, $\sin C = \sin\theta = \frac{1}{\mu} = \frac{3}{4}$
Yellow, orange and red have higher λ ⇒ lower μ ⇒ higher critical angle.
∴ These will be in emergent ray.

(122) Answer : (1)
Solution:

$\frac{1}{v} - \frac{1.5}{-120} = \frac{1-1.5}{40} = \frac{-1}{80} \;\Rightarrow\; \frac{1}{v} + \frac{1}{80} = \frac{-1}{80}$

$\Rightarrow\; \frac{1}{v} = \frac{-2}{80} = \frac{-1}{40} \;\Rightarrow\; v = -40\,\text{cm}$

(123) Answer : (1)
Solution:
Gauss law is valid in uniform electric field. If we make a gaussian surface in a uniform electric field, then net flux through it will be zero and hence net charge inside is zero.

(124) Answer : (2)
Solution:
By Lenz’s law, to decrease the magnetic flux current will decrease.

(125) Answer : (1)
Solution:
Due to symmetry, $M' = \frac{M}{4}$

(126) Answer : (2)
Solution:
M = *NIA*
= $I\pi R^2$
$\tau = \left|\overrightarrow{M} \times \overrightarrow{B}\right|$
= *MB*sin60º
$= \frac{\pi R^2 IB\sqrt{3}}{2}$

(127) Answer : (1)
Solution:
$\overrightarrow{F} = q\left(\overrightarrow{v} \times \overrightarrow{B}\right)$
$\overrightarrow{F} \neq 0$
So, the charge's velocity, path and momentum will change.

(128) Answer : (2)
Solution:
$\omega^2 = \frac{1}{LC}$
$10^4 = \frac{1}{\frac{1}{2} \times 10^2 \times 10^{-3} \times C}$
$C = \frac{2}{10^6 \times 10^{-3}}$
C = 2 mF

(129) Answer : (1)
Solution:
$\frac{N_P}{N_S} = \frac{I_S}{I_P} = \frac{1}{3}$

(130) Answer : (4)
Solution:
We know that, $h = \frac{2T\cos\theta}{R\rho g}$, if θ = 0° then liquid will rise. Liquid will not rise or fall when θ = 90°.
Venturimeter is based on Bernoulli's principle.

(131) Answer : (4)
Solution:
Since the ball is moving with constant velocity, hence the net force on ball would be zero.

(132) Answer : (1)
Solution:
$m \times s \times \Delta T = M \times L + 10 \times M$
$\Rightarrow 100 \times 1 \times 60 = M \times 540 + 10 \times M$
$\Rightarrow M = \frac{100 \times 60}{550}$
$\Rightarrow M = \frac{120}{11} = 10.9\text{ g}$

(133) Answer : (1)
Solution:
From first law of thermodynamics we know that $Q = \Delta U + W$ and if $Q = 0 \Rightarrow \Delta U + W = 0$
$\Rightarrow \Delta U = -W \Rightarrow$ process should be adiabatic.

(134) Answer : (4)
Solution:
Since, process is isobaric, therefore $V \propto T$
$W = nR\Delta T \Rightarrow W = 3 \times 2 \times (3T - T)$
= 3 × 2 × 2 × [273 + 47]
= 3 × 4 × 320
= 3840 cal

(135) Answer : (1)
Solution:
For Argon, $K.E = \frac{3}{2}k_BT$
For Oxygen, $K.E = \frac{5}{2}k_BT$
⇒ Ratio = $\frac{3}{5}$

CHEMISTRY

(136) Answer : (2)
Solution:
Solution which show large negative deviation from Raoult's law form maximum boiling azeotrope.

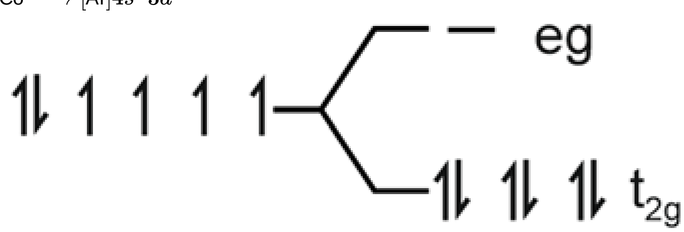
(137) Answer : (2)
Solution:
$NaCl$ → i = 1 + α = 1 + 0.9 = 1.9
$CaCl_2$ → i = 1 + 2α = 1 + 2 × 0.6 = 2.2
$AlCl_3$ → i = 1 + 3α = 1 + 3 × 0.5 = 2.5
$K_2[Fe(CN)_6]$ → i = 1 + 3α = 1 + 3 × 0.7 = 3.25

(138) Answer : (1)
Solution:
Covalent nature of ionic bond increases with increase in charge on both cation and anion and with decrease in size of cation and increase in size of anion

(139) Answer : (2)
Solution:
I_3^- → Linear (planar)
ClF_3 → Bent T shape (planar)
BF_3 → Trigonal planar
XeO_3 → Pyramidal
B_2H_6 → non-planar

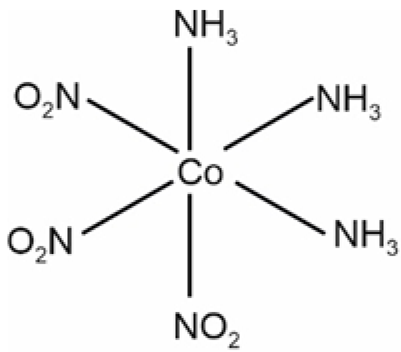
(140) Answer : (2)
Solution:
O_2 = 2
O_2^+ = 2.5
O_2^- = 1.5
O_2^{2-} = 1

(141) Answer : (2)
Solution:
CN^- ligand act as strong field ligand.
$Co^{3+} \rightarrow [Ar]4s^{\circ}3d^6$

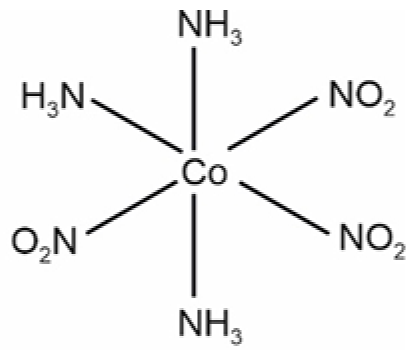


So correct configuration is $t_{2g}^6 eg^0$

(142) Answer : (1)
Solution:



Fac –

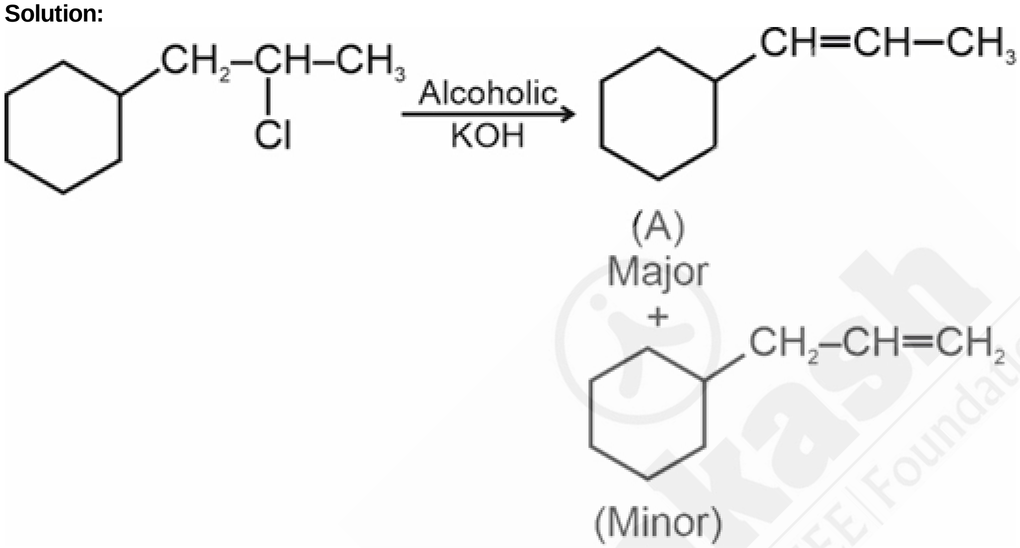


Mer –

- $[Co(en)_3]^{3+}$ complex show optical isomerism.

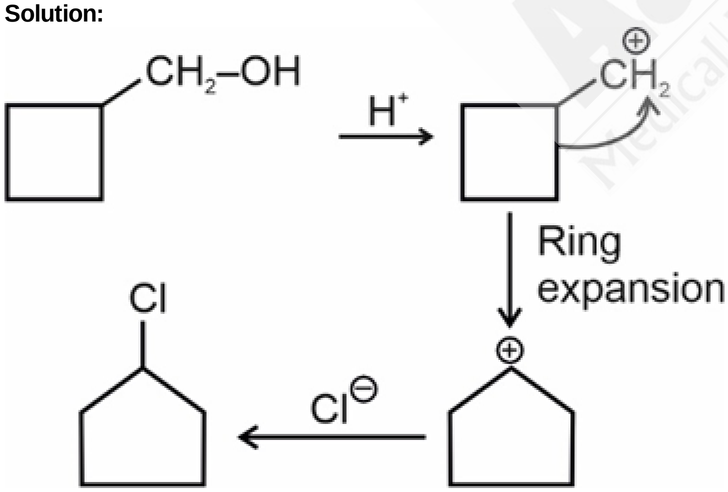
(143) Answer : (4)

Solution:



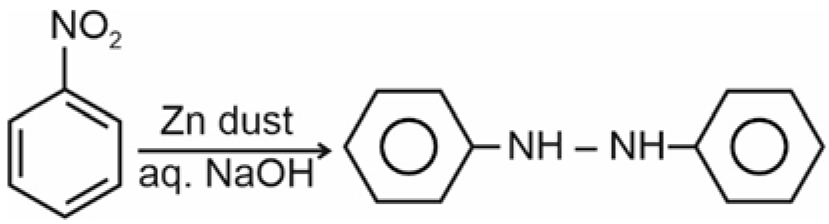
(144) Answer : (3)

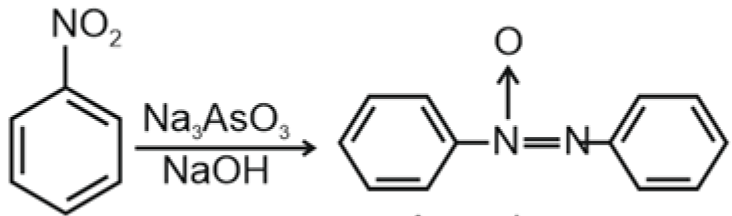
Solution:



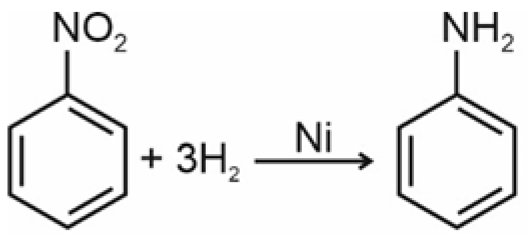
(145) Answer : (2)

Solution:

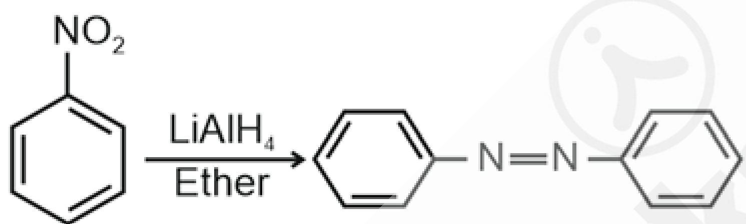




Azoxybenzene



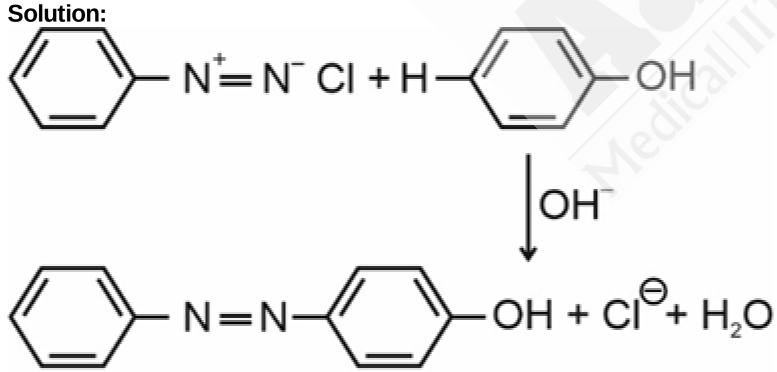
Aniline



Azobenzene

(146) Answer : (3)

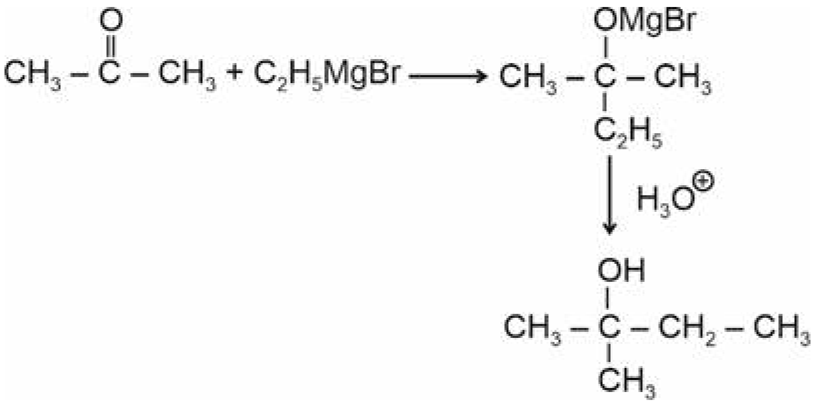
Solution:



p-Hydroxyazabenzene (orange dye)

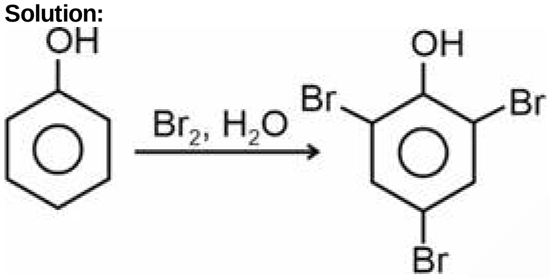
(147) Answer : (1)

Solution:



(148) Answer : (1)
Solution:
$\lambda = \frac{h}{mv}$
$\lambda \propto \frac{1}{m}$
Smaller is the molecular mass, longer is the wavelength.

(149) Answer : (3)
Solution:



(150) Answer : (4)
Solution:

Primary alcohol $\xrightarrow{KMnO_4, H^{\oplus}}$ Carboxylic acid.

(151) Answer : (1)
Solution:

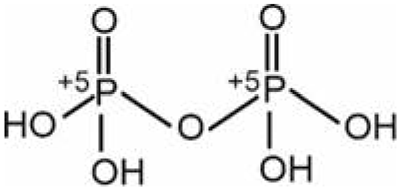
(a) $4NH_3 + 5O_2 \xrightarrow[500\,K,\,9\ bar]{Rh} 4NO + 6H_2O$ → Ostwald process

(b) $4HCl + O_2 \xrightarrow[723\,K]{CuCl_2} 2Cl_2 + 2H_2O$ → Deacon's process

(c) $N_2 + 3H_2 \xrightarrow{Fe+Mo} 2NH_3$ → Haber's process

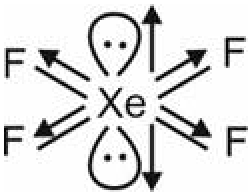
(d) $2SO_2 + O_2 \xrightarrow[673-723\,K]{V_2O_5} 2SO_3$ → Contact process

(152) Answer : (3)
Solution:



Number of P —OH bonds in $H_4P_2O_6$ = 4

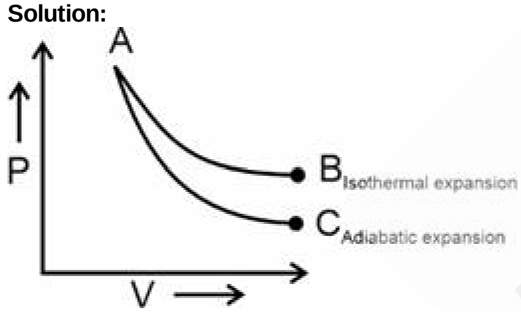
(153) Answer : (2)
Solution:
XeF_4 has zero dipole moment. It has Square planar structure due to which the bond moments of Xe – F cancel each other



(154) Answer : (2)
Solution:
La^{3+} and Lu^{3+} are diamagnetic species. Thorium exhibits only +4 oxidation state.

(155) Answer : (3)
Solution:
$Cr_2O_7^{2+} + 2OH^- \rightarrow 2CrO_4^{2-} + H_2O$
$10I^- + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 5I_2$

(156) Answer : (2)
Solution:
ΔH(kJ mol^{-1})
$\left(\frac{1}{2}A \rightarrow B\right) \times 2\ \Delta H = -200$
$3B \rightarrow 2C + D \quad \Delta H = +120$
$2D \rightarrow E + A \quad \Delta H = +325$
ΔH for B + D → E + 2C
= (–200 + 120 + 325) kJ mol^{-1}
= +245 kJ mol^{-1}

(157) Answer : (2)
Solution:


During adiabatic expansion, cooling takes place.

(158) Answer : (1)
Solution:
Since density of water 1 g/mL
∴ mass of water in 36 mL = 36 g
Moles = 2 moles

(159) Answer : (3)
Solution:

(Atomic number)	(Symbol)
102	Unb
115	Uup
114	Uuq
107	Uns

(160) Answer : (4)
Solution:
Higher is the reduction potential, greater is the oxidising power,
So, order of reducing power will be : C > B > A

(161) Answer : (2)
Solution:
$\Delta G = -RT\ln K_C$

$= -8.314 \times 300 \times \ln(2 \times 10^{13})$
$\Delta G = -7.64 \times 10^{4}$ J mol^{-1}

(162) Answer : (2)
Solution:
$F^{-} = 1s^2 2s^2 2p^6$ i.e. (Ne)
$Zn^{2+} = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$)
$Na^{+} = 1s^2 2s^2 2p^6$ i.e. (Ne)
$Ca^{2+} = 1s^2 2s^2 2p^6 3s^2 3p^6$ i.e. (Ar)

(163) Answer : (2)
Solution:
$\alpha = \frac{\Lambda_m}{\Lambda_m{}^{\infty}}$
$\alpha = \frac{10}{250} = 0.04$
As α is quite less; α <<< 1
Hence, it is negligible
$K_a = c\alpha^2 = 0.02(0.04)^2$
$= 3.2 \times 10^{-5}$

(164) Answer : (1)
Solution:

$$\begin{array}{lcccccc} & CH_3\,COONa & + & HCl & \longrightarrow & CH_3\,COOH & + & NaCl \\ t=0 & 100 \times 0.1 & & 50 \times 0.1 & & 0 & & \\ t=t & 5 & & 5 & & 5 & & \end{array}$$

Now solution consists of CH_3COOH and CH_3 COONa i.e. weak acid and salt of weak acid and strong base form acidic buffer solution.
$S_8(s) + 8O_2(g) \rightleftharpoons 8SO_2(g)$, $K_P = K_C$.

(165) Answer : (1)
Solution:
Order of first ionisation enthalpy :
Si > Ge > Pb > Sn

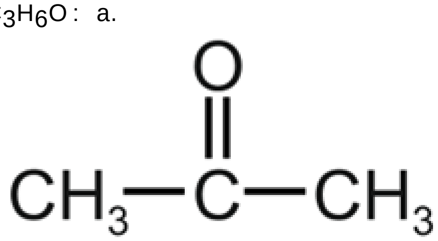
(166) Answer : (4)
Solution:
The reaction in which same element gets oxidised as well as reduced is known as disproportionation reaction.

(i) $\overset{0}{P}_4 + 3\,OH^- \rightarrow \overset{3}{P}H_3 + 3H_2\overset{+1}{P}O_2^-$

(ii) $2\overset{+1}{H}_2\overset{-1}{O}_2 \rightarrow 2H_2\overset{-2}{O} + \overset{0}{O}_2$

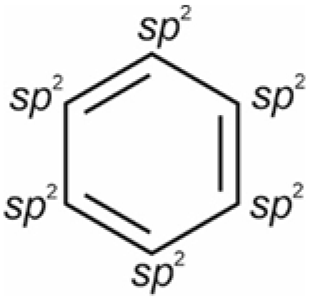
(iii) $\overset{0}{Cl}_2 + 2\,OH^- \rightarrow \overset{+1}{Cl}O^- + \overset{-1}{Cl^-} + H_2O$
All are disproportionation reactions

(167) Answer : (3)
Solution:
C_3H_6O :   a.



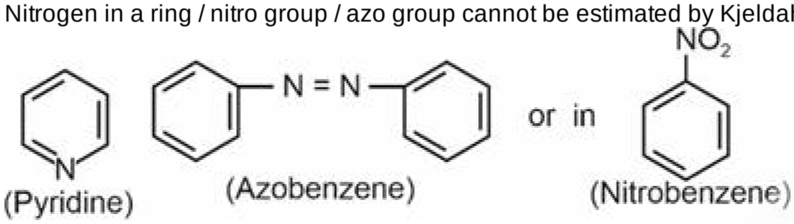
cannot show metamerism.

b. $H_3C - \overset{\oplus}{C}H_2$ has 3α hydrogen and it is stabilised by hyperconjugation while in $\overset{\oplus}{C}H_3$ there is no possibility of hyperconjugation.
c. Hyperconjugation is a permanent effect.
d.



All the carbon atoms are sp^2 hybridised.

(168) Answer : (1)
Solution:
Nitrogen in a ring / nitro group / azo group cannot be estimated by Kjeldahl method.



Nitrogen cannot be estimated by Kjeldahl method

(169) Answer : (2)
Solution:
The correct priority order is
–COOH > $-SO_3H$ > –COOR > –CN

(170) Answer : (2)
Solution:
Hydrolysis of cane sugar in acidic medium is a pseudo first order reaction.

(171) Answer : (4)
Solution:
$\log \frac{k_1}{k_2} = \frac{Ea}{2.303\,R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

$\log 2 = \frac{Ea}{2.303 \times 8.314}\left(\frac{1}{300} - \frac{1}{310}\right)$

Ea = 0.3 × 2.303 × 8.314 × 300 × 31 J
= 53.42 kJ

(172) Answer : (1)
Solution:

Acid	**pKa**
CF_3COOH	0.23
C_6H_5COOH	4.19
CH_3COOH	4.76
C_6H_5COOH	10

(173) Answer : (4)
Solution:
CH_3CHO will give yellow precipitate with I_2 + NaOH but C_6H_5CHO will not give yellow precipitate.

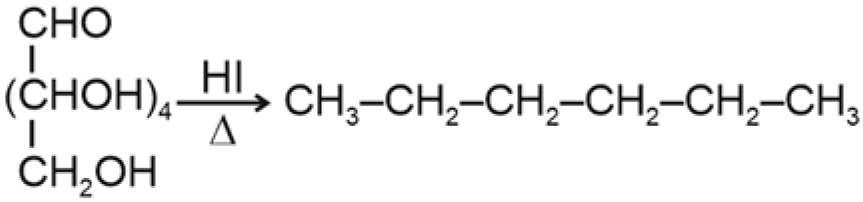
(174) Answer : (2)
Solution:
Borax bead is $NaBO_2 + B_2O_3$.

(175) Answer : (4)
Solution:
Na^+, Mg^{2+} and Ca^{2+} belongs to group V cations.

(176) Answer : (1)

Solution:
Electron donating group attached with benzene ring increases the rate of electrophilic substitution reaction.

(177) Answer : (3)
Solution:
Glucose in an aldohexose and dextrorotatory compound



(178) Answer : (1)
Solution:

Sol.: $\frac{1}{\lambda} = \mathrm{R_H}\left(\frac{1}{\mathrm{n}_1^2} - \frac{1}{\mathrm{n}_2^2}\right)$

$\Rightarrow\ \frac{1}{\lambda} = \mathrm{R_H}\left(\frac{1}{1^2} - \frac{1}{2^2}\right)$

$\Rightarrow\ \frac{1}{\lambda} = \mathrm{R_H}\left(\frac{4-1}{4}\right)$

$\Rightarrow\ \lambda = \frac{4}{3\mathrm{R_h}} = \frac{4}{3\times 109677}\,\mathrm{cm}$

$\Rightarrow\ \lambda = 1.2157\times 10^{-5}\mathrm{cm} = 1.2157\times 10^{-7}\,\mathrm{cm}$

$\Rightarrow\ \lambda = 1.2157\times 10^{-9}\mathrm{cm} \simeq 121.6\ \mathrm{cm}$

(179) Answer : (4)
Solution:
The secondary structure of protein refers to the shape in which a long polypeptide chain each exist. They are found to exist in two different types of structures viz. α-helix and β-pleated sheet structure.

(180) Answer : (4)
Solution: